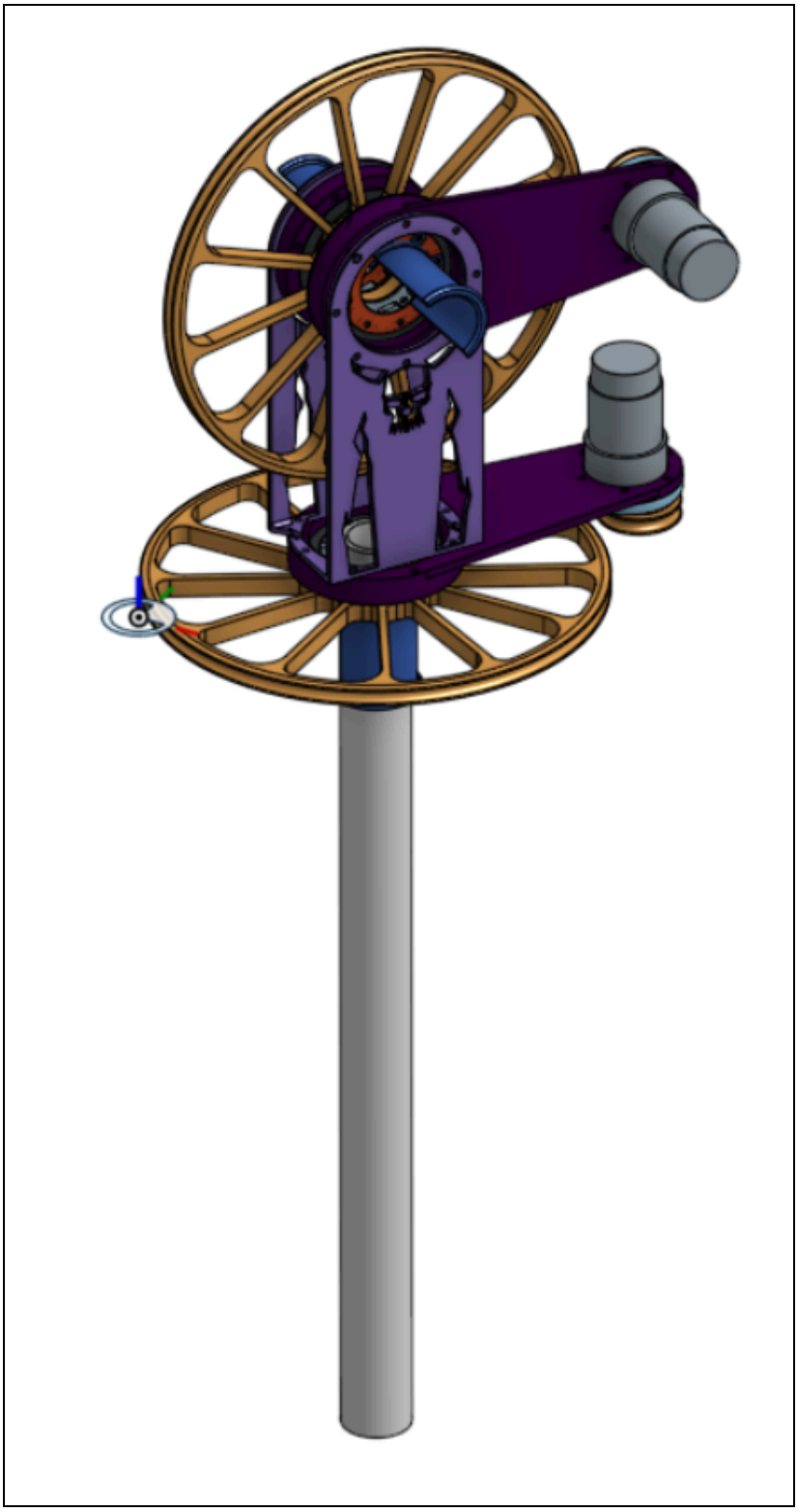
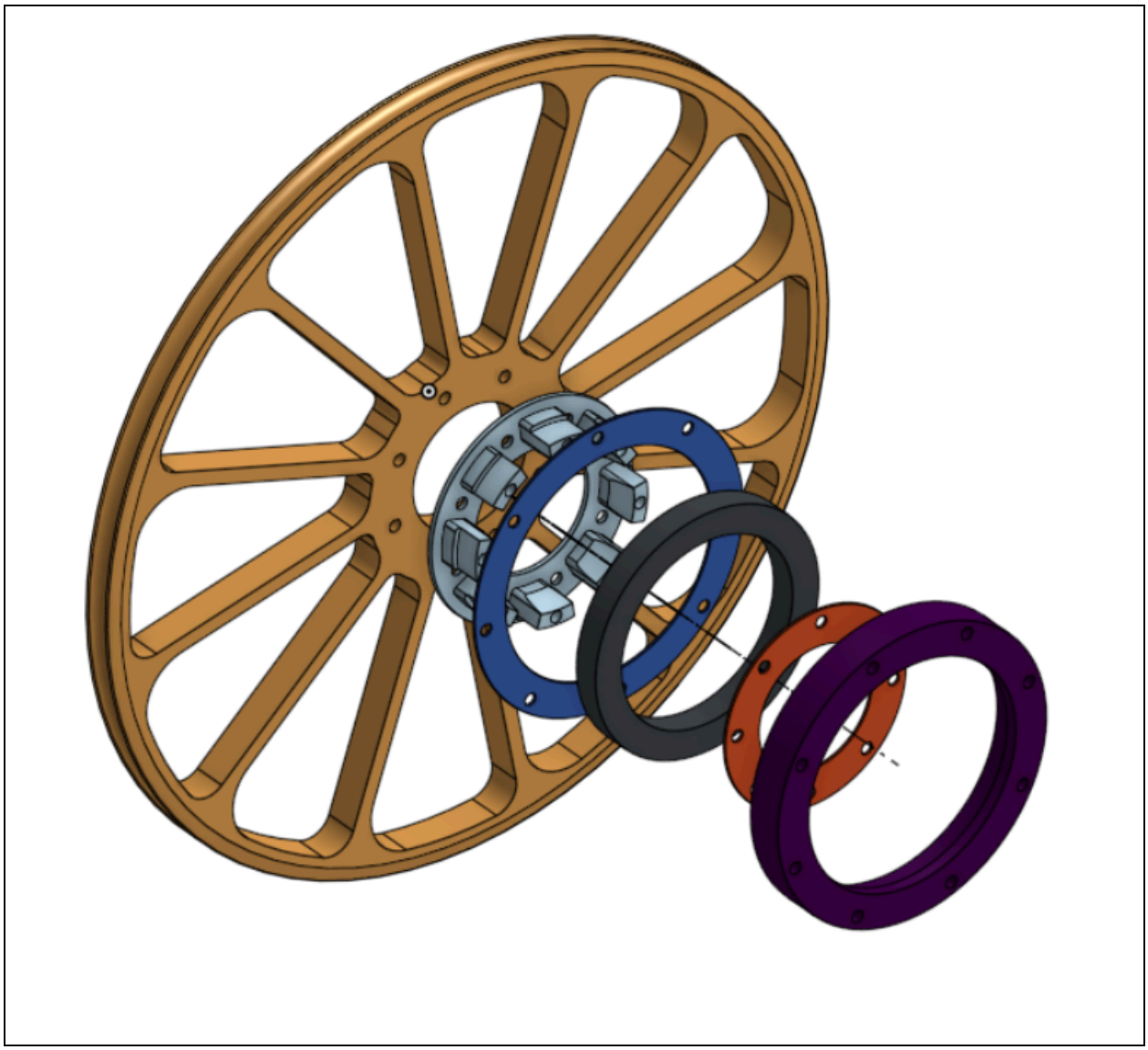


MiniPos Belt Driven Az-El Rotator with [6.667:1 = 20:3] Gear Ratio



Bearing Assembly Part Order is as Such (Right To Left)
Part Name: [8” Pulley Wheel - Bearing Hub -Bearing Retainer L - Bearing - Bearing Retainer S - Bearing Collar]
Part #:[2- 8 - 11 - 7 - 10 - 9]

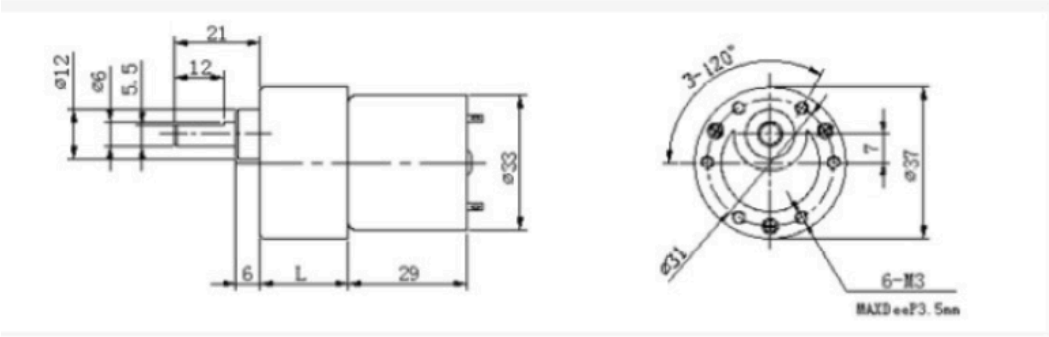
BOM

Part #	Name	Section	Quantity	Procurment	Cost ($)	3D Print Mass (g)
1	1.2" Pully Wheel	Pully Drive	2	FDM	-	5
2	8" Pully Wheel	Pully Drive	2	FDM	-	78
3	Polyeurethane Belt	Pully Drive	2	Amazon	10	
4	Motor Carrier	Pully Drive	2	FDM	-	50
5	Motor Collar	Pully Drive	2	FDM	-	7
6	Geared Motor	Pully Drive	2	Digikey	33	
7	50x65mm Bearing	Bearing Assy.	3	Amazon	20	
8	Bearing Hub	Bearing Assy.	3	FDM	-	8
9	Bearing Collar	Bearing Assy.	1	FDM	-	17
10	Bearing Retainer S	Bearing Assy.	3	FDM	-	2
11	Bearing Retainer L	Bearing Assy.	3	FDM	-	2
12	Upright Mounts	Structure	1	FDM	-	35
13	Shaft Coupler	Structure	3	FDM	-	8
14	12V PSU	Power	1	Amazon	20	
15	12-5V Step Down	Power	1	Amazon	12	
16	Motor Controller	Control	1	Amazon	25	
17	ESP32	Control	1	Digikey	10	

Approximate Cost TOtal ~$200
Approximate FDM Mass ~400g

Technical Specs and Drawings

Metal DC Geared Motor w/Encoder - 12V 83RPM [45Kg.cm]

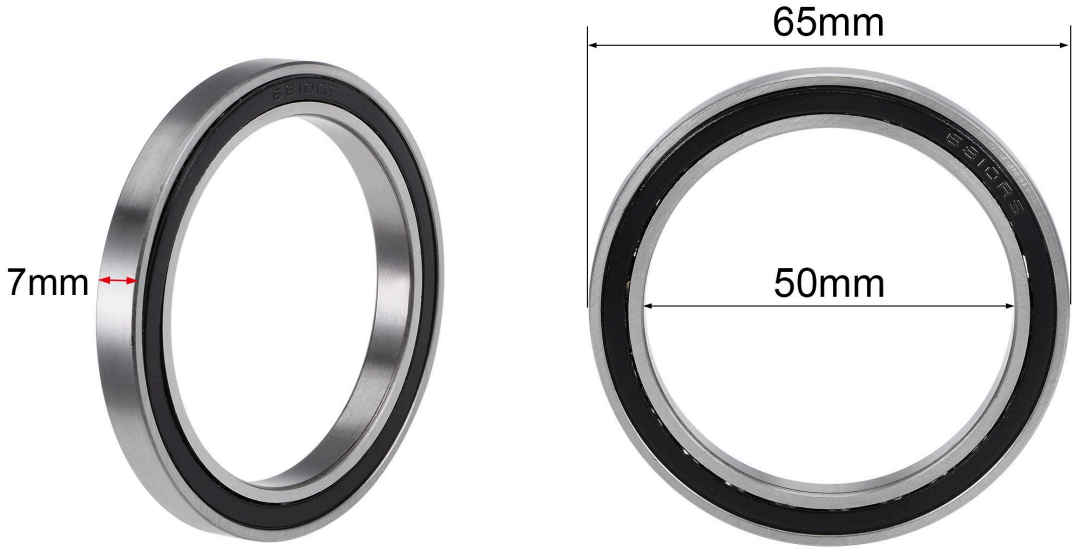


上图未加编码器尺寸，编码器长度10MM
L为减速箱尺寸，和实际减速比相关

The encoder makes the motor 12mm longer
The value of L is 24mm

Gear ratio: 131:1 No-load speed: 83 + 10% RPM No-load current: 350 mA Start Voltage: 1.0 V Stall Torque: 45 Kg.com Stall Current: 7 A Insulation resistance: 20 M Ω EncoderOperating Voltage: 5 V Encoder type: Hall Encoder Resolution: 16CPR(motor shaft)/2096CPR(gearbox shaft) Weight: 205g

Ball Bearings - Deep Groove Ball Bearings



[PolyUrethane Belt]
Rule of Thumb, Cut the Belt 5-10% shorter than the loop to maintain tension.

[Power Supply] 120vAC to 12v DC

Step Down Converter 12v to 5v

Motor Controllers

ESP 32

Thank you! Your order has been submitted.

Your registration has been submitted.

Reference Information

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WEB ID
372382780

SUBMITTED
05/31/2026 17:35:24

Address

SHIPPING ADDRESS
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Colorado State University
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APT B
FORT COLLINS, CO 80521-6028
USA

EMAIL
connerharte@gmail.com

PHONE
19706188554

CUSTOMER TYPE
School/University

Shipping

SHIPPED BY
DigiKey Electronics
Shipping from USA

SHIP METHOD
FedEx Ground

SALESORDER ID
99528300

Payment

BILLING ADDRESS
Conner Harte
Colorado State University
715 GLENMOOR DR
APT B
FORT COLLINS, CO 80521-6028
USA

PAYMENT METHOD
Visa | Ending in 7314

All prices are in USD

#	PRODUCT DETAILS	QUANTITY	AVAILABILITY	UNIT PRICE	EXTENDED PRICE
1	1965-ESP32-DEVKITC-32UE-ND ESP32-DEVKITC-32UE ESP32-WROOM-32UE 4MB FLASH	3	Immediate	10.00000	$30.00
2	1738-1105-ND FIT0185 GEARMOTOR 83 RPM 12V W/ENCODER	2	Immediate	16.50000	$33.00

These commodities, technology or software will be exported from the United States in accordance with the Export Administration regulation. These products may not be re-exported, diverted, or otherwise shipped to another destination that does not fully comply with U.S. laws and the laws of any other applicable jurisdiction.

Order Details

Order placed May 31, 2026 | Order # 114-8801788-4426631 Invoice

Ship to
Conner Harte
715 GLENMOOR DR APT B
FORT COLLINS, CO 80521-6028
United States

Payment method
VISA Visa ending in 7314
View related transactions

Order Summary

Item(s) Subtotal:	$122.03
Shipping & Handling:	$6.99
Free Shipping:	-$6.99
Total before tax:	$122.03
Estimated tax to be collected:	$10.12
CO Retail Delivery Fees	$0.28
Grand Total:	**$132.43**

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ALITOVE DC 12V 10A 120W Power Supply Adapter Transformer Switch AC 110V / 220V to DC 12V 10amp Switching Converter LED Driver for LED Strip Light CCTV Camera Security System
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uxcell 6810-2RS Deep Groove Ball Bearings 50mm Inner Dia 65mm OD 7mm Bore Double Sealed Chrome Steel Z2 2pcs
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